Problems Set

Statistical Methods

August 12, 2026

Problem 1

A national logistics company offers three tiers of shipping for its retail partners: Overnight, Standard, and Economy. Analyzing their annual shipping volume, the company notes that 20% of all packages are shipped Overnight, 50% are shipped Standard, and 30% are shipped Economy.

Despite their best efforts, packages are sometimes delayed. Their performance metrics indicate that the probability of a package arriving late is 5% for Overnight, 15% for Standard, and 30% for Economy.

(a) What is the overall proportion of packages that arrive late in a given year?

(b) Suppose a customer leaves a review stating their package arrived *on time* (it was not delayed). What is the probability that they paid for Overnight shipping?

(c) Under the same condition (the package arrived on time), what is the probability that they used Standard shipping?

Problem 2

When the exact dependencies between events are unknown, it is often impossible to calculate the exact probability of their intersection. However, **Bonferroni's Inequality** provides a guaranteed lower bound.
For two events A and B, the inequality states:

$$P(A \cap B) \geq P(A) + P(B) - 1$$

(a) **The Base Proof:** Use the fundamental axioms of probability (specifically the general addition rule for the union of two events) to rigorously prove the inequality stated above.

(b) **Extension to Three Events:** Using your result from part (a), derive a similar lower bound for the intersection of three events, $P(A \cap B \cap C)$.

(c) **Application:** A data center relies on three critical infrastructure subsystems to maintain uptime: Power (E_1), Cooling (E_2), and Networking (E_3). The individual reliabilities over one year are $P(E_1) = 0.96$, $P(E_2) = 0.92$, and $P(E_3) = 0.88$. The failure dependencies between them are completely unknown. Using your derived formula from part (b), calculate the absolute minimum probability that *all three* subsystems function correctly throughout the year.

Problem 3

A red die, a blue die, and a yellow die (all standard six-sided dice) are rolled. We are interested in the probability that the number appearing on the blue die is less than the number appearing on the yellow die, which is in turn less than the number appearing on the red die.
Let B, Y, and R be the numbers appearing on the blue, yellow, and red dice, respectively. We want to analyze the event $B < Y < R$.

(a) What is the probability that no two of the dice land on the same number?

(b) Given that no two of the dice land on the same number, what is the conditional probability that $B < Y < R$?

(c) What is the unconditional probability $P(B < Y < R)$?

Problem 4

A closet contains 10 distinct pairs of shoes (20 shoes total). Suppose you are in a rush and select 8 shoes completely at random in the dark.

(a) What is the probability that your selection contains *exactly zero* complete pairs of shoes?

(b) What is the probability that your selection contains *exactly one* complete pair of shoes?

(c) Let E be the event that you select at least one complete pair, and F be the event that you select exactly two complete pairs. Calculate the conditional probability $P(F \mid E)$.

Problem 5

Consider an electrical circuit with three critical relays, denoted A, B, and C. The probability of each relay failing during a power surge is $P(A) = P(B) = P(C) = \frac{1}{2}$.
The relays are designed such that they fail pairwise independently; that is, the failure of any one relay does not affect the probability of failure of any *single* other relay. However, if relays A and B both fail simultaneously, the resulting electrical load guarantees that relay C will also fail (i.e., $P(C \mid A \cap B) = 1$).

(a) Calculate the probability that all three relays fail, $P(A \cap B \cap C)$.

(b) Are the events A, B, and C mutually independent? Use the formal definition of mutual independence to rigorously prove your answer.

(c) What is the probability that *exactly one* relay fails during a power surge? *(Hint: A Venn diagram or partition of the sample space may be highly useful here.)*

Problem 6

You are presented with two opaque urns. Urn 1 initially contains 4 red marbles and 6 black marbles. Urn 2 initially contains 5 red marbles and 5 black marbles. The experiment proceeds in two distinct stages:

1. Two marbles are drawn simultaneously at random from Urn 1 and transferred to Urn 2. Their colors are not observed.
2. The contents of Urn 2 are thoroughly mixed, and a single marble is drawn from Urn 2.

(a) Use the Law of Total Probability to find the overall probability that the final marble drawn from Urn 2 is red.

(b) Suppose the final marble drawn from Urn 2 is revealed to be red. Given this information, what is the exact probability that the two marbles transferred from Urn 1 were both black?

(c) Under the same condition (the final marble drawn from Urn 2 is red), what is the probability that the transferred marbles consisted of exactly one red and one black marble?

Problem 7: Pólya's Urn Scheme

An urn initially contains 2 red marbles and 1 blue marble. A marble is drawn at random from the urn, and its color is observed. It is then returned to the urn along with **one additional marble of the exact same color**. This process (drawing, observing, and adding a duplicate) is repeated over multiple trials.

Let R_k and B_k denote the events that a red marble or a blue marble is drawn on the k-th trial, respectively.

(a) **Marginal Probability:** Calculate the unconditional probability that the second marble drawn is red, $P(R_2)$. How does this compare to $P(R_1)$, and what does this imply about the marginal distribution of any single draw in this process?

(b) **Retroactive Inference (Bayes' Theorem):** Suppose you step out of the room for the first draw and only observe the *second* draw, which happens to be blue. Given this information, what is the probability that the first marble drawn was red?

(c) **Exchangeability:** Calculate the probability of observing the exact sequence (Red, Red, Blue) across the first three draws. Next, calculate the probability of observing the sequence (Blue, Red, Red). Compare your results. What fundamental property of the Pólya urn model does this illustrate regarding sequences of the same composition but different order?

Problem 8: Pairwise Independence Does Not Imply Mutual Independence

Consider a regular tetrahedron (a four-sided die) where the four faces are distinctly painted. Face 1 is painted entirely Red, Face 2 is entirely Green, and Face 3 is entirely Blue. The fourth face, Face 4, is painted with a mixture of all three colors: Red, Green, and Blue. The die is rolled once in a fair manner, such that each of the four faces is equally likely to land face down (the "outcome" face). Let R, G, and B be the events that the outcome face contains the colors Red, Green, and Blue, respectively.

(a) **Marginal Probabilities:** Calculate the individual probabilities $P(R)$, $P(G)$, and $P(B)$.

(b) **Pairwise Independence:** Prove that the events R, G, and B are pairwise independent. You must explicitly calculate $P(R \cap G)$, $P(R \cap B)$, and $P(G \cap B)$ and demonstrate that they equal the product of their respective marginal probabilities.

(c) **Mutual Independence Test:** Calculate the probability of the intersection of all three events, $P(R \cap G \cap B)$.

(d) **Conclusion:** Does $P(R \cap G \cap B) = P(R)P(G)P(B)$? Briefly explain why this geometric counterexample proves that a collection of events being pairwise independent is an insufficient condition for them to be mutually independent.

Problem 9: Simpson's Paradox and Confounding Variables

A research hospital is testing a new experimental treatment (Treatment A) against the current standard of care (Treatment B) for a specific disease. The overall success of the treatments is recorded, but the researchers also stratify the data based on the initial severity of the disease (Mild vs. Severe).

The recovery data is summarized in the following tables:

Mild Cases:

	Recovered	Did Not Recover	Total
Treatment A	90	10	100
Treatment B	80	20	100

Severe Cases:

	Recovered	Did Not Recover	Total
Treatment A	120	280	400
Treatment B	20	80	100

Let R be the event of recovery, T_A and T_B be the events of receiving Treatment A and Treatment B, and M and S be the events of having a Mild or Severe case, respectively.

(a) **Aggregated Analysis:** Calculate the overall probability of recovery for each treatment, $P(R \mid T_A)$ and $P(R \mid T_B)$, by pooling the mild and severe data together. Based *only* on this aggregated data, which treatment appears more effective?

(b) **Stratified Analysis:** Calculate the conditional probabilities of recovery given the severity for each treatment: $P(R \mid T_A \cap M)$, $P(R \mid T_B \cap M)$, $P(R \mid T_A \cap S)$, and $P(R \mid T_B \cap S)$. Which treatment is more effective for Mild cases? Which is more effective for Severe cases?

(c) **The Paradox:** The contradiction observed between parts (a) and (b) is known as Simpson's Paradox. Using the Law of Total Probability, express the overall rate $P(R \mid T_A)$ in terms of the stratified probabilities and the marginal allocation probabilities $P(M \mid T_A)$ and $P(S \mid T_A)$.

(d) **Conceptual Explanation:** Explain both mathematically and conceptually how Treatment A can be superior in every individual subgroup, yet appear inferior overall. What role does the unequal sample allocation (the confounding variable) play here?